PUBLIC TRANSPORT USAGE ANALYZER

Course End Project 9 — Batch-9 | Transportation

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1. Problem Statement

Analyze public transport usage to identify high-demand routes, low-utilization services and peak travel periods, and use the findings to support better service allocation.

2. Objectives

• Store and analyze route and passenger records.
• Calculate route-wise passenger counts.
• Detect peak travel hours.
• Identify high-demand and underutilized routes.
• Store processed records in SQLite DBMS.
• Provide an interactive visualization dashboard.
• Extend the solution with ML-based passenger-demand prediction.

3. Dataset

The selected online dataset is the Public Transportation Usage Dataset surfaced through Kaggle. Its published description reports 569,400 hourly observations, 88 columns, six cities, multiple transport modes, and features including boarding count, occupancy and delay. The source describes it as a GTFS-inspired simulated transit dataset and states a CC0 1.0 license.

Source: https://www.kaggle.com/code/amith1707/eda-transits-utilities-demo/input

4. Technology Stack

Python, Pandas, NumPy, SQLite, Scikit-learn, Plotly and Streamlit.

5. System Workflow

Dataset → validation/cleaning → feature preparation → SQLite storage → exploratory analysis → dashboard visualization → Random Forest demand prediction → recommendations.

6. Implementation

The implementation contains separate modules for demo-data generation, DBMS creation, analysis and ML prediction. The dashboard provides filters for city, mode and route, KPI cards, hourly demand trends, top-route rankings, daily trends and utilization comparisons.

7. Demonstration Results

Metric	Demonstration result
Records analyzed	97,920
Peak average-demand hour	08:00
High-utilization observations	48.64%
ML model	Random Forest Regressor
Test MAE	15.46 passengers
Test R^2	0.8799

Top Demonstration Routes

TOK-07 (Tokyo metro), TOK-08 (Tokyo rail), NEW-07 (New York metro), TOK-06 (Tokyo bus), and NEW-08 (New York rail) were the highest-volume routes in the demonstration data.

8. Dashboard

Run `streamlit run dashboard/app.py` from the project folder. The dashboard is designed to satisfy the course requirement for practical data visualization and route/period analysis.

9. Recommendations

• Increase service frequency around detected peak periods.
• Prioritize high-demand routes for additional capacity.
• Review low-utilization services for schedule optimization.
• Use demand forecasts as a planning aid rather than as a sole operational decision source.
• Refresh the dashboard with the full source dataset before final submission.

10. Limitations

The repository includes a reproducible demonstration subset rather than the full ~840 MB source file. The source itself is described as simulated/GTFS-inspired. For the final academic submission, download the source dataset, map its columns to the project schema, rerun the analysis, and update the numerical results in the report.

11. GitHub Repository

Paste the final GitHub repository URL here after uploading the project.